[SoC Design]

Week 5

Report 5

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1. Simulation Results

hy_soc.v

```
module hy_soc (
  input wire clk,     // system clock
  input wire rst_n,    // async assert, sync deassert (active-low)
    output wire [3:0] led
);
  // ---------------------------------------------------------------------------
  // AHB-Lite bus signals
  // ---------------------------------------------------------------------------
  (* DONT_TOUCH = "TRUE" *) wire [31:0] haddr;
  (* DONT_TOUCH = "TRUE" *) wire [31:0] hwdata;
  (* DONT_TOUCH = "TRUE" *) wire        hwrite;
  (* DONT_TOUCH = "TRUE" *) wire [1:0]  htrans;
  wire [2:0]  hburst;
  wire        hmastlock;
  wire [3:0]  hprot;
  wire [2:0]  hsize;
  wire [31:0] hrdata;
  wire        hresp;
  (* DONT_TOUCH = "TRUE" *) wire        hready;
  // Subordinate signals
  wire        hsel_s0;
  wire [31:0] hrdata_s0;
  wire        hreadyout_s0;
  // No interrupts in v01
  wire [31:0] irq = 32'h0;
  // ---------------------------------------------------------------------------
  // Reset synchronizer: two independent async-assert / sync-deassert chains
  //   por_sync (ext reset only)        -> PORESETn, DBGRESETn
  //   sys_sync (ext reset + SYSRESETREQ) -> HRESETn
  // ---------------------------------------------------------------------------
  wire hresetn, poresetn, dbgresetn;
  cm0_rst_sync u_rst_sync (
    .clk            (clk),
    .rst_n          (rst_n),
    .sysresetreq_i  (1'b0),
    .poresetn_o     (poresetn),
    .dbgresetn_o    (dbgresetn),
    .hresetn_o      (hresetn)
  );
  // CDBGPWRUP self-ack
  wire cdbgpwrupreq;
  // ---------------------------------------------------------------------------
  // Cortex-M0 DesignStart
  // ---------------------------------------------------------------------------
  CORTEXM0INTEGRATION u_cm0 (
    // clocks & resets
    .FCLK           (clk),
    .SCLK           (clk),
    .HCLK           (clk),
    .DCLK           (clk),
    .PORESETn       (poresetn),
    .DBGRESETn      (dbgresetn),
    .HRESETn        (hresetn),
    .SWCLKTCK       (1'b0),
    .nTRST          (1'b1),
    // AHB-Lite manager port
    .HADDR          (haddr),
    .HBURST         (hburst),
    .HMASTLOCK      (hmastlock),
    .HPROT          (hprot),
    .HSIZE          (hsize),
    .HTRANS         (htrans),
```

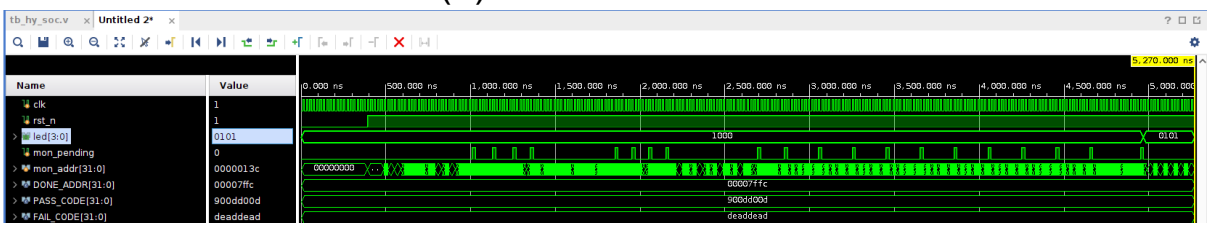
```
    .HWDATA          (hwdata),
    .HWRITE          (hwrite),
    .HRDATA          (hrdata),
    .HREADY          (hready),
    .HRESP           (hresp),
    .HMASTER         (),
    // code sequentiality (unused)
    .CODENSEQ        (),
    .CODEHINTDE      (),
    .SPECHTRANS      (),
    // debug — SWD tied off
    .SWDITMS         (1'b0),
    .TDI             (1'b0),
    .SWDO            (),
    .SWDOEN          (),
    .TDO             (),
    .nTDOEN          (),
    .DBGRESTART      (1'b0),
    .DBGRESTARTED    (),
    .EDBGRQ          (1'b0),
    .HALTED          (),
    // interrupts
    .NMI             (1'b0),
    .IRQ             (irq),
    .TXEV            (),
    .RXEV            (1'b0),
    .LOCKUP          (),
    .SYSRESETREQ     (),
    // SysTick: 50 MHz → 10 ms = 500 000 - 1 = 0x07_A11F
    .STCALIB         ({1'b1, 1'b0, 24'h07A11F}),
    .STCLKEN         (1'b0),
    .IRQLATENCY      (8'h00),
    .ECOREVNUM       (28'h0),
    // power management — tied off
    .GATEHCLK        (),
    .SLEEPING        (),
    .SLEEPDEEP       (),
    .WAKEUP          (),
    .WICSENSE        (),
    .SLEEPHOLDREQn   (1'b1),
    .SLEEPHOLDACKn   (),
    .WICENREQ        (1'b0),
    .WICENACK        (),
    .CDBGPWRUPREQ    (cdbgpwrupreq),
    .CDBGPWRUPACK    (cdbgpwrupreq),  // self-ack
    // scan — tied off
    .SE              (1'b0),
    .RSTBYPASS       (1'b0)
  );
  // ---------------------------------------------------------------------------
  // On-chip Interconnect (single subordinate, no default slave)
  // ---------------------------------------------------------------------------
  ahb_interconnect u_interconnect (
    .hclk            (clk),
    .hrst_n          (hresetn),
    .haddr_i         (haddr),
    .htrans_i        (htrans),
    .hrdata_o        (hrdata),
    .hready_o        (hready),
    .hresp_o         (hresp),
    .hsel_sram_o     (hsel_s0),
    .hrdata_sram_i   (hrdata_s0),
    .hreadyout_sram_i(hreadyout_s0)
  );
  // ---------------------------------------------------------------------------
  // s0: ahb_sram 64 KB (MEMWIDTH=16 → 2^16 bytes)
```

```
  // ---------------------------------------------------------------------------
  ahb_sram #(.MEMWIDTH(16)) u_sram (
    .hsel      (hsel_s0),
    .clk       (clk),
    .rst_n     (hresetn),
    .hready    (hready),
    .haddr     (haddr),
    .htrans    (htrans),
    .hwrite    (hwrite),
    .hsize     (hsize),
    .hwdata    (hwdata),
    .hreadyout (hreadyout_s0),
    .hrdata    (hrdata_s0)
  );
    localparam DONE_ADDR = 32'h00007FFC;
    localparam PASS_CODE = 32'h900DD00D;
    localparam FAIL_CODE = 32'hDEADDEAD;
    reg        mon_pending;
    reg [31:0] mon_addr;

// state
    reg [3:0] c_led, n_led;

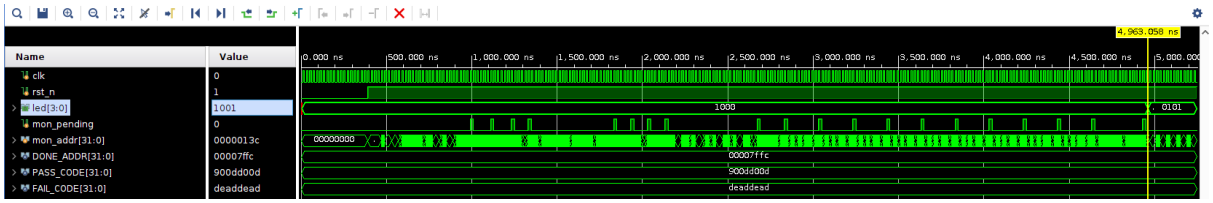
    assign led = c_led;
    always @(posedge clk or negedge hresetn) begin
        if (!hresetn) begin
            mon_pending <= 1'b0;
            mon_addr    <= 32'h0;
        end else if (hready) begin
            mon_pending <= hwrite & htrans[1];
            mon_addr    <= haddr;
        end
    end
    always @(posedge clk or negedge hresetn) begin
        if (!hresetn) begin
            c_led <= 4'b1000;
            n_led <= 4'b1100;
        end else begin
            if (mon_pending && mon_addr == DONE_ADDR) begin
                if      (hwdata == PASS_CODE) begin
                    c_led <= 4'b0101;
                end
                else if (hwdata == FAIL_CODE) begin
                    c_led <= 4'b1101;
                end
            end
       end
    end
endmodule
```

Behave simulation result(1)

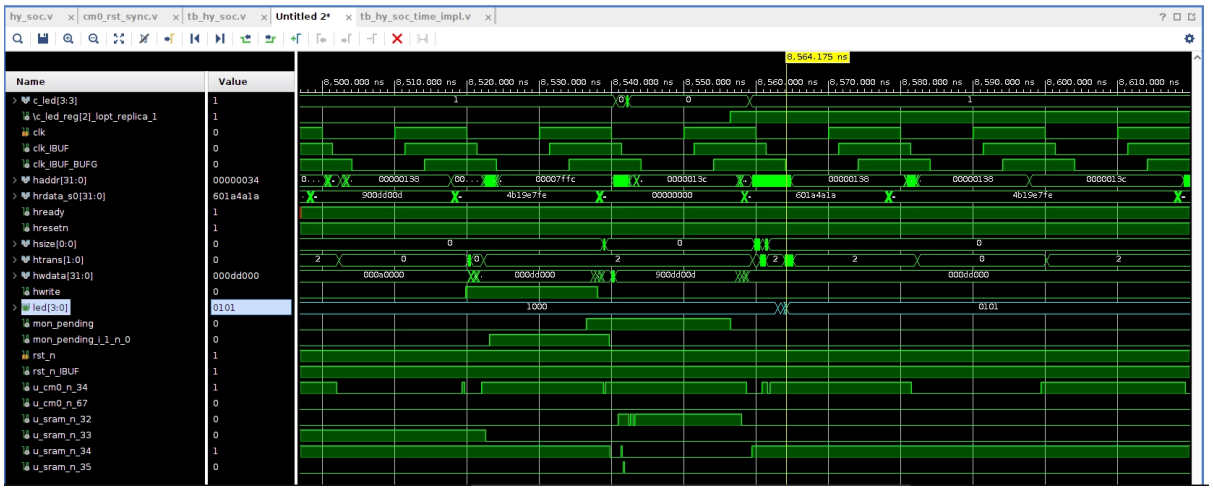


Behave simulation result(2)

Post-implementation simulation result (1)



Post-implementation simulation result (2)



Terminal result

```
# run 1000ns
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_hy_soc_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:09 ; elapsed = 00:00:09 . Memory (MB): peak = 10146.117 ; gain = 12.469 ; free physical = 28665 ; free virtual = 38234
run all
TEST PASS! (t=9290000 ns)
$finish called at time : 9290 ns : File "/home/c0rnpotato/codespace/soc/soc2/soc_week4.srcs/sim_1/imports/tb/tb_hy_soc.v" Line 68
```

2. Overview

The errors that previously occurred in Keil have been resolved, and the Hex file was successfully generated. This Hex file contains a firmware image that accesses __main via startup_cm0.S to execute test.c, and test.c verifies the functional integrity of the hardware through PASS_CODE. In Vivado, the testbench monitoring and pass/fail detection logic was modified and mapped to the LED pins. Appropriate XDC constraints were written to ensure correct port mapping for the LED and asynchronous reset. Additionally, simulations performed after implementation confirmed that the design operates normally without any asynchronous reset issues.

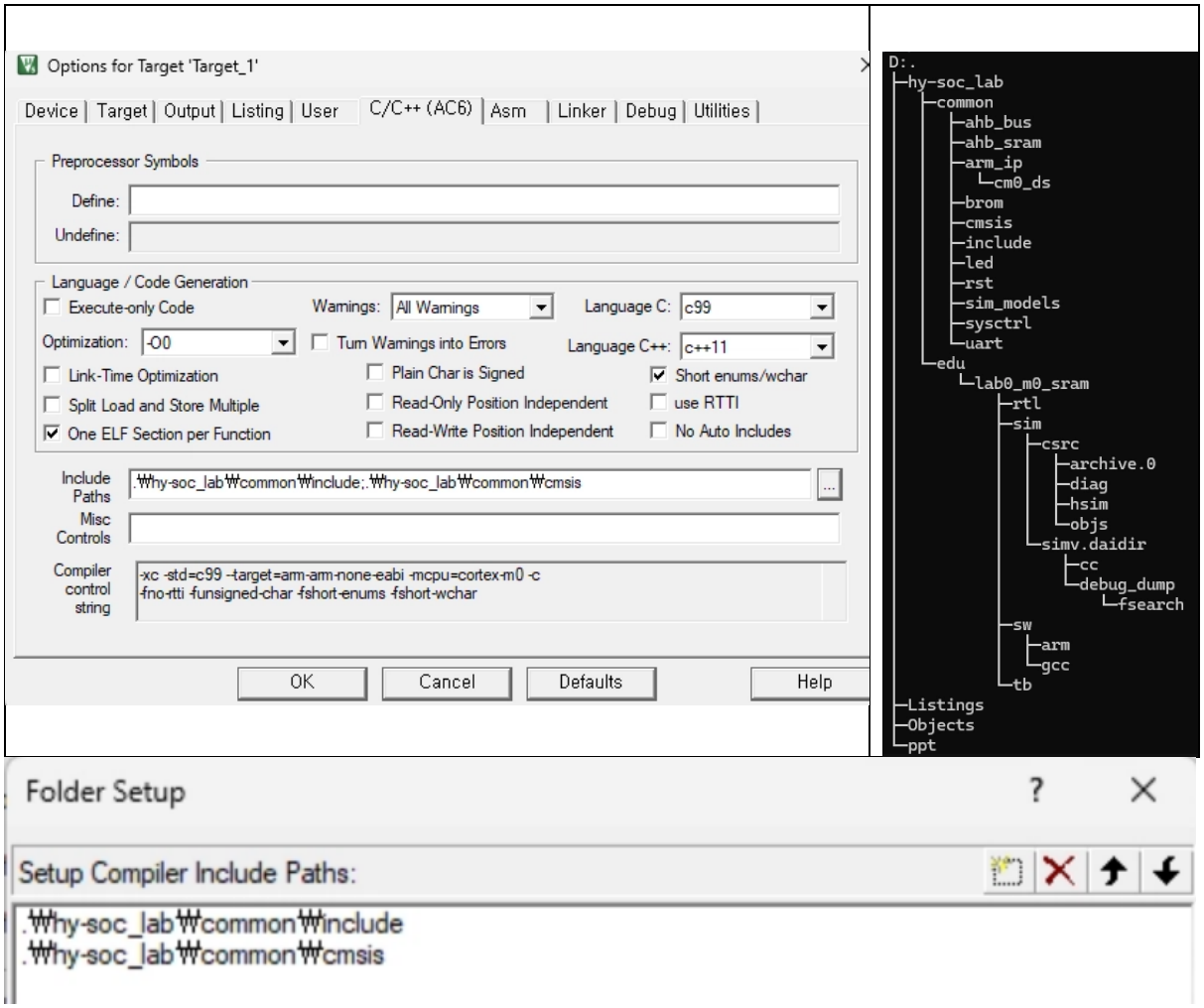
Cortex M0 and BRARM were integrated via the AAHB bus, and async reset logic was combined to enable external initialization. Furthermore, firmware was written in Keil to extract the Hex file, and by porting it, the successful integration of the SoC was confirmed.

3. Details

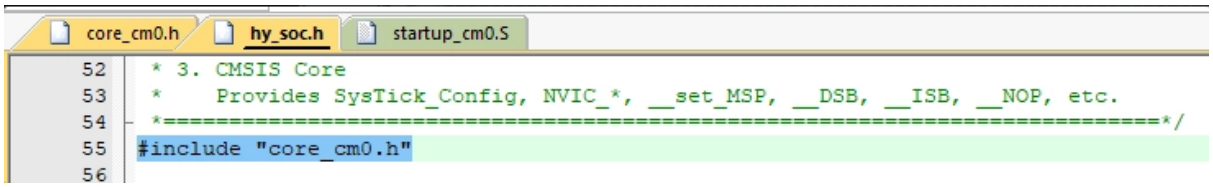
i) Keil

a. Include path

The initial documentation contained path recognition issues. To address this, the teaching assistant (TA) recommended converting all directories to absolute paths. However, this approach presented significant drawbacks regarding maintainability. To resolve this more effectively, the 'Include Paths' configuration within the Keil environment was utilized to add the two specific directories listed below, thereby ensuring both functional resolution and project portability.



All include directives within the source code omit explicit path specifications, as demonstrated below.



b. Linker Warning

Although the Hex file is successfully generated after resolving the path issues, the following warning message persists:

The root cause of this issue lies in a discrepancy of operational granularity; while the conventional linker operates on an 8-bit basis, the current configuration specifies operations in 4-bit units as described below.

```
/* ---------- Reset Handler ---------- */
    .section .text
    .p2align 2
    .thumb_func
    .globl  Reset_Handler
    .type   Reset_Handler, %function
```

Consequently, the EDA tool issues a warning to signal a potential conflict. However, as this configuration is intentional, the warning can be addressed by applying the specific reserved keyword found in the official documentation to indicate that the code is 'by design.' In practice, both Hex files were verified to function correctly regardless of the warning. The Hex file exhibiting the warning is denoted as code_gen1.mem, while the version with the resolution implemented was tested as code_gen2.mem.

https://kolegite.com/EE_library/standards/ARM_ABI/addenda32.pdf

```
    3  Reserved
    n  (in 4..12) Code was permitted to depend on the 8-byte alignment of 8-byte data items
       and the alignment of data items having up to 2n-byte extended alignment

Tag_ABI_align_preserved, (=25), uleb128
    0  The user did not require code to preserve 8-byte alignment of 8-byte data objects
    1  Code was required to preserve 8-byte alignment of 8-byte data objects
    2  Code was required to preserve 8-byte alignment of 8-byte data objects and to
       ensure (SP MOD 8) = 0 at all instruction boundaries (not just at function calls)
    3  Reserved
    n (in 4..12) Code was required to preserve the alignments of case 2 and the alignment
      of data items having up to 2n-byte extended alignment.
```

startup_cm0.S

```
/*
 * startup_cm0.S — Cortex-M0 startup for HY-SoC Lab0 (Arm Compiler 6 / armclang)
 *
 * Minimal startup: Reset_Handler + Default_Handler only.
 * No SysTick or named exception handlers needed.
 */
    .syntax unified
    .cpu cortex-m0
    .thumb
    .eabi_attribute 25, 1
```

Build Output

```
Rebuild started: Project: week5
*** Using Compiler 'V6.24', folder: 'E:\MDK\ARM\ARMCLANG\Bin'
Rebuild target 'Target_1'
Note: source file '.\hy-soc_lab\edu\lab0_m0_sram\sw\arm\startup_cm0.S' - object file renamed from '.\Objects\startup_cm0.o' to '.\Objects\startup_cm0_1.o'.
Note: source file '.\hy-soc_lab\edu\lab0_m0_sram\sw\arm\startup_cm0.S' - object file renamed from '.\Objects\startup_cm0.o' to '.\Objects\startup_cm0_1.o'.
compiling test.c...
assembling startup_cm0.S...
linking...
Program Size: Code=372 RO-data=128 RW-data=0 ZI-data=2048
After Build - User command #1: fromelf --vhx --32x1 --output=code.hex ./Objects/week5.axf
".\Objects\week5.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed:  00:00:02
```

Although a discrepancy exists between the generated binary values and the provided reference files, functional integrity was confirmed as the 'PASS_CODE' was successfully triggered within the Vivado environment.

ii) Vivado

a. hex file path problem (code.hex → code.mem)

Vivado does not recognize .hex files as valid design sources. To address this compatibility issue, the file extension was modified to .mem for the implementation process. Furthermore, it was confirmed that the Hex files generated by Keil operate consistently and exhibit identical behavior under the same modification, as previously discussed.

```
// Memory array
reg [31:0] mem [0:(2**(MEMWIDTH-2)-1)];

initial begin
  //$readmemh("code.hex", mem);
  $readmemh("code.mem", mem);
  //$readmemh("code_gen1.mem", mem);
  //$readmemh("code_gen2.mem", mem);
end
```

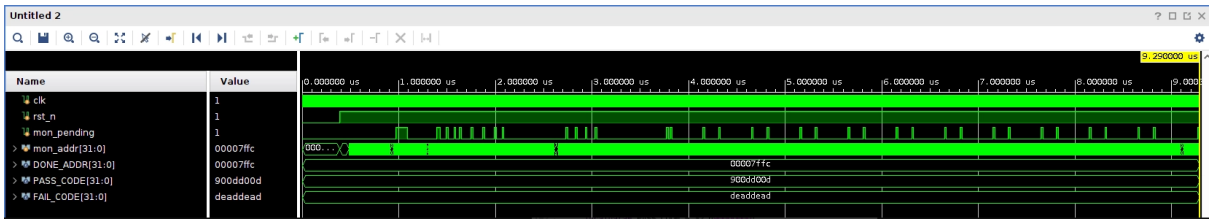
```
// Memory array
reg [31:0] mem [0:(2**(MEMWIDTH-2)-1)];

initial begin
  //$readmemh("code.hex", mem);
  //$readmemh("code.mem", mem);
  $readmemh("code_gen1.mem", mem);
  //$readmemh("code_gen2.mem", mem);
end
```

```
// Memory array
reg [31:0] mem [0:(2**(MEMWIDTH-2)-1)];

initial begin
  //$readmemh("code.hex", mem);
  //$readmemh("code.mem", mem);
  //$readmemh("code_gen1.mem", mem);
  $readmemh("code_gen2.mem", mem);
end
```

```
# run 1000ns
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_hy_soc_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:09 ; elapsed = 00:00:09 . Memory (MB): peak = 10146.117 ; gain = 12.469 ; free physical = 28665 ; free virtual = 38234
run all
TEST PASS! (t=9290000 ns)
$finish called at time : 9290 ns : File "/home/c0rnpotato/codespace/soc/soc2/soc_week4.srcs/sim_1/imports/tb/tb_hy_soc.v" Line 68
```



b. HIGH-Z problem

In the behavioral simulation of the existing design, High-Z (High Impedance) states were observed during operation. This issue originates from accessing the BRAM while it contains uninitialized or 'garbage' values. Although the initial access is intended for a write operation, the process of fetching a word to perform an overwrite causes indeterminate data to be automatically driven onto the hrdata channel. Nevertheless, as the BRAM is currently the only slave device connected to the bus, this behavior is expected to have a negligible impact on system integrity at this stage.

c. synthesis & implementation

```
    localparam DONE_ADDR = 32'h00007FFC;
    localparam PASS_CODE = 32'h900DD00D;
    localparam FAIL_CODE = 32'hDEADDEAD;

    reg        mon_pending;
    reg [31:0] mon_addr;


// state
    reg [3:0] c_led, n_led;

    assign led = c_led;

    always @(posedge clk or negedge hresetn) begin
        if (!hresetn) begin
            mon_pending <= 1'b0;
            mon_addr    <= 32'h0;
        end else if (hready) begin
            mon_pending <= hwrite & htrans[1];
            mon_addr    <= haddr;
        end
    end

    always @(posedge clk or negedge hresetn) begin
        if (!hresetn) begin
            c_led <= 4'b1000;
            n_led <= 4'b1100;
        end else begin
            if (mon_pending && mon_addr == DONE_ADDR) begin
                if      (hwdata == PASS_CODE) begin
                    c_led <= 4'b0101;
                end
                else if (hwdata == FAIL_CODE) begin
                    c_led <= 4'b1101;
                end
            end
        end
    end
```

```
module hy_soc {
  input wire clk,      // system clock
  input wire rst_n,     // async assert, sync deassert (active-low)
  output wire [3:0] led
};

  // ---------------------------------------------
  // AHB-Lite bus signals
  // ---------------------------------------------
  (* DONT_TOUCH = "TRUE" *) wire [31:0] haddr;
  (* DONT_TOUCH = "TRUE" *) wire [31:0] hwdata;
  (* DONT_TOUCH = "TRUE" *) wire        hwrite;
  (* DONT_TOUCH = "TRUE" *) wire [1:0]  htrans;
  wire [2:0]  hburst;
  wire        hmastlock;
  wire [3:0]  hprot;
  wire [2:0]  hsize;
  wire [31:0] hrdata;
  wire        hresp;
  (* DONT_TOUCH = "TRUE" *) wire        hready;

  // Subordinate signals
  wire        hsel_s0;
  wire [31:0] hrdata_s0;
  wire        hreadyout_s0;

  // No interrupts in v01
  wire [31:0] irq = 32'h0;
```

An LED output port was implemented for FPGA deployment, along with the corresponding control logic. This logic was developed by adapting and tailoring specific segments of the testbench (TB) code. While n_led was present in the skeleton code, it was excluded from the final design as it was redundant; instead, c_led was utilized exclusively. For debugging purposes, specific LED states were defined: 4'b1000 indicates the post-reset state, and 4'b0101 represents the state following successful bootloading. Although physical validation on the FPGA has not yet been finalized, the simulation results predict successful hardware operation as demonstrated below.

Furthermore, the rst_n for this logic was synchronized with hresetn. Previously, using an alternative reset signal led to a discrepancy where the design passed behavioral simulation but triggered a FAIL_CODE during post-implementation simulation. This synchronization issue is identified as the likely root cause of the malfunctions observed during initial FPGA testing.

d. timing margin problem

Timing — **Setup** | Hold | Pulse Width

Worst Negative Slack (WNS):	-6.055 ns
Total Negative Slack (TNS):	-3834.909 ns
Number of Failing Endpoints:	1423
Total Number of Endpoints:	2430

Implemented Timing Report

Operating the system at 100MHz resulted in timing violations, as illustrated in the following results. To resolve these timing issues, the clock constraints in the XDC file were adjusted to reduce the frequency to 50MHz. Furthermore, physical pin assignments were defined for the reset and LED signals within the XDC file to ensure proper hardware mapping.

```
#Clock signal
set_property -dict { PACKAGE_PIN K17   IOSTANDARD LVCMOS33 } [get_ports { clk }]; #IO_L12P_T1_MRCC_35 Sch=sysclk
#create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 4} [get_ports { clk }];
create_clock -add -name sys_clk_pin -period 20.00 -waveform {0 10} [get_ports { clk }];

##Switches
#set_property -dict { PACKAGE_PIN G15   IOSTANDARD LVCMOS33 } [get_ports { sw[0] }]; #IO_L19N_T3_VREF_35 Sch=sw[0]
#set_property -dict { PACKAGE_PIN P15   IOSTANDARD LVCMOS33 } [get_ports { sw[1] }]; #IO_L24P_T3_34 Sch=sw[1]
#set_property -dict { PACKAGE_PIN W13   IOSTANDARD LVCMOS33 } [get_ports { sw[2] }]; #IO_L4N_T0_34 Sch=sw[2]
#set_property -dict { PACKAGE_PIN T16   IOSTANDARD LVCMOS33 } [get_ports { sw[3] }]; #IO_L9P_T1_DQS_34 Sch=sw[3]


##Buttons
set_property -dict { PACKAGE_PIN K18   IOSTANDARD LVCMOS33 } [get_ports { rst_n }]; #IO_L12N_T1_MRCC_35 Sch=btn[0]
#set_property -dict { PACKAGE_PIN P16   IOSTANDARD LVCMOS33 } [get_ports { btn[1] }]; #IO_L24N_T3_34 Sch=btn[1]
#set_property -dict { PACKAGE_PIN K19   IOSTANDARD LVCMOS33 } [get_ports { btn[2] }]; #IO_L10P_T1_AD11P_35 Sch=btn[2]
#set_property -dict { PACKAGE_PIN Y16   IOSTANDARD LVCMOS33 } [get_ports { btn[3] }]; #IO_L7P_T1_34 Sch=btn[3]


#LEDs
set_property -dict { PACKAGE_PIN M14   IOSTANDARD LVCMOS33 } [get_ports { led[0] }]; #IO_L23P_T3_35 Sch=led[0]
set_property -dict { PACKAGE_PIN M15   IOSTANDARD LVCMOS33 } [get_ports { led[1] }]; #IO_L23N_T3_35 Sch=led[1]
set_property -dict { PACKAGE_PIN G14   IOSTANDARD LVCMOS33 } [get_ports { led[2] }]; #IO_0_35 Sch=led[2]
set_property -dict { PACKAGE_PIN D18   IOSTANDARD LVCMOS33 } [get_ports { led[3] }]; #IO_L3N_T0_DQS_AD1N_35 Sch=led[3]
```

The results are presented below, confirming the successful completion of the implementation process,

Project Summary × hy_soc.v × setting.xdc ×

Overview | Dashboard

Settings Edit

Project name:	soc_week4
Project location:	/home/c0mpotato/codespace/soc/soc2
Product family:	Zynq-7000
Project part:	xc7z020clg400-1
Top module name:	hy_soc
Target language:	Verilog
Simulator language:	Mixed
Target Simulator:	Vivado Simulator

Synthesis

Status:	✓ Complete
Messages:	! 50 warnings
Part:	xc7z020clg400-1
Strategy:	Vivado Synthesis Defaults
Report Strategy:	Vivado Synthesis Default Reports
Incremental synthesis:	Automatically selected checkpoint

Implementation **Summary** | Route Status

Status:	✓ Complete
Messages:	! 1 warning
Part:	xc7z020clg400-1
Strategy:	Vivado Implementation Defaults
Report Strategy:	Vivado Implementation Default Reports
Incremental implementation:	None

DRC Violations

Summary: ! 34 warnings

Implemented DRC Report

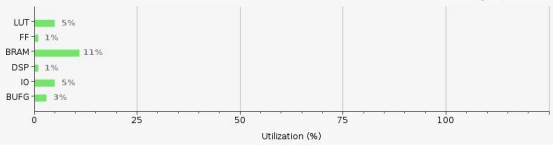
Timing **Setup** | Hold | Pulse Width

Worst Negative Slack (WNS):	0.661 ns
Total Negative Slack (TNS):	0 ns
Number of Failing Endpoints:	0
Total Number of Endpoints:	2396

Implemented Timing Report

Utilization Post-Synthesis | **Post-Implementation**

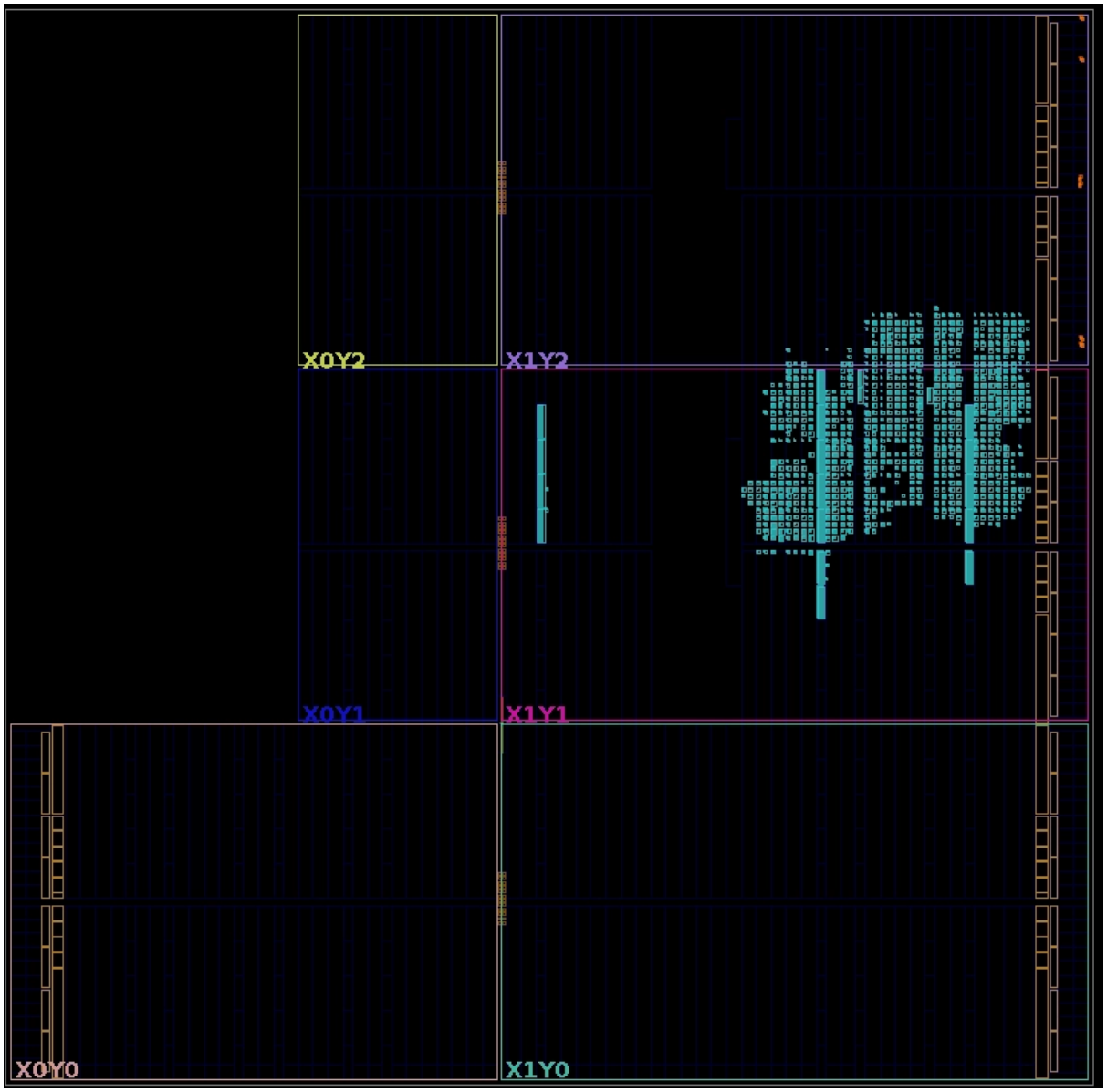
Graph | Table



Power **Summary** | On-Chip

Total On-Chip Power:	**0.151 W**
Junction Temperature:	**26.7 °C**
Thermal Margin:	58.3 °C (4.9 W)
Effective ϑJA:	11.5 °C/W
Power supplied to off-chip devices:	0 W
Confidence level:	Medium

Implemented Power Report



iii) Source code

a. keil

Except for startup_cm0.S, I used the provided code exactly as is without modification. Regarding the path issue, I set the include path and used it by entering only the filename.

startup_cm0.S

```
/*
 * startup_cm0.S — Cortex-M0 startup for HY-SoC Lab0 (Arm Compiler 6 / armclang)
 *
 * Minimal startup: Reset_Handler + Default_Handler only.
 * No SysTick or named exception handlers needed.
 */
    .syntax unified
    .cpu cortex-m0
```

```
    .thumb
         .eabi_attribute 25, 1
/* ---------- Stack top (matches scatter.sct ARM_LIB_STACK) ---------- */
    .equ    __initial_sp, 0x00010000    /* SRAM 64 KB top */
    .globl  __initial_sp
/* ---------- Vector Table (RESET area) ---------- */
    .section RESET, "a", %progbits
    .p2align 3
    .globl  __Vectors
    .globl  __Vectors_End
    .globl  __Vectors_Size
__Vectors:
    .word   __initial_sp        /* 0x00: Initial Stack Pointer          */
    .word   Reset_Handler       /* 0x04: Reset                          */
    .word   Default_Handler     /* 0x08: NMI                            */
    .word   Default_Handler     /* 0x0C: Hard Fault                     */
    .rept   12
    .word   Default_Handler     /* 0x10-0x3C: reserved + SysTick        */
    .endr
    .rept   16
    .word   Default_Handler     /* 0x40-0x7C: IRQ0-15                   */
    .endr
__Vectors_End:
    .equ __Vectors_Size, __Vectors_End - __Vectors
/* ---------- Reset Handler ---------- */
    .section .text
    .p2align 2
    .thumb_func
    .globl  Reset_Handler
    .type   Reset_Handler, %function
Reset_Handler:
    ldr     r0, =__main         /* Arm C library: scatter-load + __rt_entry → main() */
    bx      r0
/* ---------- Default Handler (weak) ---------- */
    .thumb_func
    .weak   Default_Handler
    .type   Default_Handler, %function
Default_Handler:
    b       .
    .end
```

code.hex (generated)

```
00010000
000000B1
000000B5
000000B5
000000B5
000000B5
000000B5
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000000B5
000000B5
000000B5
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000000B5
46C046C0
F804F000
BD1FB51F
BD10B510
46854803
FFF8F7FF
F828F000
F80EF000
00010000
F7FFB403
BC03FFF1
F810F000
47004801
0000E7FE
00000081
4604B510
46C046C0
F7FF4620
BD10FFED
00004770
20184901
E7FEBEAB
00020026
48024943
E7FF6008
46C0E7FE
DEADDEAD
B084B580
90032000
030A2101
03499202
9A029101
60114937
4A379B02
9B02605A
609A4A36
60D09A02
68009802
D0034288
F7FFE7FF
E7FFFFDD
68409802
4288492E
E7FFD003
FFD4F7FF
9802E7FF
492B6880
D0034288
F7FFE7FF
E7FFFFCB
20119901
99017008
70482022
20339901
99017088
70C82044

```
78009801
D0032811
F7FFE7FF
E7FFFFB7
78409801
D0032822
F7FFE7FF
E7FFFFAF
78809801
D0032833
F7FFE7FF
E7FFFFA7
78C09801
D0032844
F7FFE7FF
E7FFFF9F
20009902
900060C8
9800E7FF
DC092809
9902E7FF
1C4068C8
E7FF60C8
1C409800
E7F29000
68C09802
D003280A
F7FFE7FF
E7FFFF85
48064905
E7FF6008
46C0E7FE
A5A5A5A5
12345678
DEADBEEF
00007FFC
900DD00D
```

b. vivado

The XDC constraint file was integrated, and the hy_soc.v module was updated to include the LED control and monitor logic. All other source files remain unmodified to ensure the integrity of the original design.

hy_soc.v

```
// --=========================================================================--
// Copyright (c) 2025-2026 DSAL, Hanyang University. All rights reserved
//                     DSAL Confidential Proprietary
//  ---------------------------------------------------------------------------
//  This confidential and proprietary software may be used only as
//  authorised by a licensing agreement from DSAL.
// ---------------------------------------------------------------------------
// FILE NAME       : hy_soc.v
// DEPARTMENT      : Digital System Architecture Lab.
// AUTHOR          : Ji-Hoon Kim
// AUTHOR'S EMAIL  : jhoonkim@hanyang.ac.kr
// ---------------------------------------------------------------------------
```

```
// hy_soc.v — HY-SoC Educational Lab0
//
// Cortex-M0 (DesignStart) + AHB-Lite Interconnect + SRAM 64KB
//
// Memory map:
//   0x0000_0000 – 0xFFFF_FFFF : single subordinate — ahb_sram (64 KB)
//   (no default slave; unmapped address handling introduced in lab1)
`timescale 1ns/1ps
module hy_soc (
  input wire clk,     // system clock
  input wire rst_n,    // async assert, sync deassert (active-low)
    output wire [3:0] led
);
  // ---------------------------------------------------------------------------
  // AHB-Lite bus signals
  // ---------------------------------------------------------------------------
  (* DONT_TOUCH = "TRUE" *) wire [31:0] haddr;
  (* DONT_TOUCH = "TRUE" *) wire [31:0] hwdata;
  (* DONT_TOUCH = "TRUE" *) wire        hwrite;
  (* DONT_TOUCH = "TRUE" *) wire [1:0]  htrans;
  wire [2:0]  hburst;
  wire        hmastlock;
  wire [3:0]  hprot;
  wire [2:0]  hsize;
  wire [31:0] hrdata;
  wire        hresp;
  (* DONT_TOUCH = "TRUE" *) wire        hready;
  // Subordinate signals
  wire        hsel_s0;
  wire [31:0] hrdata_s0;
  wire        hreadyout_s0;
  // No interrupts in v01
  wire [31:0] irq = 32'h0;
  // ---------------------------------------------------------------------------
  // Reset synchronizer: two independent async-assert / sync-deassert chains
  //   por_sync (ext reset only)        -> PORESETn, DBGRESETn
  //   sys_sync (ext reset + SYSRESETREQ) -> HRESETn
  // ---------------------------------------------------------------------------
  wire hresetn, poresetn, dbgresetn;
  cm0_rst_sync u_rst_sync (
    .clk            (clk),
    .rst_n          (rst_n),
    .sysresetreq_i  (1'b0),
    .poresetn_o     (poresetn),
    .dbgresetn_o    (dbgresetn),
    .hresetn_o      (hresetn)
  );
  // CDBGPWRUP self-ack
  wire cdbgpwrupreq;
  // ---------------------------------------------------------------------------
  // Cortex-M0 DesignStart
  // ---------------------------------------------------------------------------
  CORTEXM0INTEGRATION u_cm0 (
    // clocks & resets
    .FCLK           (clk),
    .SCLK           (clk),
    .HCLK           (clk),
    .DCLK           (clk),
    .PORESETn       (poresetn),
    .DBGRESETn      (dbgresetn),
    .HRESETn        (hresetn),
    .SWCLKTCK       (1'b0),
    .nTRST          (1'b1),
    // AHB-Lite manager port
    .HADDR          (haddr),
    .HBURST         (hburst),
```

```
    .HMASTLOCK      (hmastlock),
    .HPROT          (hprot),
    .HSIZE          (hsize),
    .HTRANS         (htrans),
    .HWDATA         (hwdata),
    .HWRITE         (hwrite),
    .HRDATA         (hrdata),
    .HREADY         (hready),
    .HRESP          (hresp),
    .HMASTER        (),
    // code sequentiality (unused)
    .CODENSEQ       (),
    .CODEHINTDE     (),
    .SPECHTRANS     (),
    // debug — SWD tied off
    .SWDITMS        (1'b0),
    .TDI            (1'b0),
    .SWDO           (),
    .SWDOEN         (),
    .TDO            (),
    .nTDOEN         (),
    .DBGRESTART     (1'b0),
    .DBGRESTARTED   (),
    .EDBGRQ         (1'b0),
    .HALTED         (),
    // interrupts
    .NMI            (1'b0),
    .IRQ            (irq),
    .TXEV           (),
    .RXEV           (1'b0),
    .LOCKUP         (),
    .SYSRESETREQ    (),
    // SysTick: 50 MHz → 10 ms = 500 000 - 1 = 0x07_A11F
    .STCALIB        ({1'b1, 1'b0, 24'h07A11F}),
    .STCLKEN        (1'b0),
    .IRQLATENCY     (8'h00),
    .ECOREVNUM      (28'h0),
    // power management — tied off
    .GATEHCLK       (),
    .SLEEPING       (),
    .SLEEPDEEP      (),
    .WAKEUP         (),
    .WICSENSE       (),
    .SLEEPHOLDREQn  (1'b1),
    .SLEEPHOLDACKn  (),
    .WICENREQ       (1'b0),
    .WICENACK       (),
    .CDBGPWRUPREQ   (cdbgpwrupreq),
    .CDBGPWRUPACK   (cdbgpwrupreq),  // self-ack
    // scan — tied off
    .SE             (1'b0),
    .RSTBYPASS      (1'b0)
  );
  // ---------------------------------------------------------------------------
  // On-chip Interconnect (single subordinate, no default slave)
  // ---------------------------------------------------------------------------
  ahb_interconnect u_interconnect (
    .hclk            (clk),
    .hrst_n          (hresetn),
    .haddr_i         (haddr),
    .htrans_i        (htrans),
    .hrdata_o        (hrdata),
    .hready_o        (hready),
    .hresp_o         (hresp),
    .hsel_sram_o     (hsel_s0),
    .hrdata_sram_i   (hrdata_s0),
```

```
    .hreadyout_sram_i(hreadyout_s0)
  );
  // ---------------------------------------------------------------------------
  // s0: ahb_sram 64 KB (MEMWIDTH=16 → 2^16 bytes)
  // ---------------------------------------------------------------------------
  ahb_sram #(.MEMWIDTH(16)) u_sram (
    .hsel      (hsel_s0),
    .clk       (clk),
    .rst_n     (hresetn),
    .hready    (hready),
    .haddr     (haddr),
    .htrans    (htrans),
    .hwrite    (hwrite),
    .hsize     (hsize),
    .hwdata    (hwdata),
    .hreadyout (hreadyout_s0),
    .hrdata    (hrdata_s0)
  );
    localparam DONE_ADDR = 32'h00007FFC;
    localparam PASS_CODE = 32'h900DD00D;
    localparam FAIL_CODE = 32'hDEADDEAD;
    reg        mon_pending;
    reg [31:0] mon_addr;

// state
    reg [3:0] c_led, n_led;

    assign led = c_led;
    always @(posedge clk or negedge hresetn) begin
        if (!hresetn) begin
            mon_pending <= 1'b0;
            mon_addr    <= 32'h0;
        end else if (hready) begin
            mon_pending <= hwrite & htrans[1];
            mon_addr    <= haddr;
        end
    end
    always @(posedge clk or negedge hresetn) begin
        if (!hresetn) begin
            c_led <= 4'b1000;
            n_led <= 4'b1100;
        end else begin
            if (mon_pending && mon_addr == DONE_ADDR) begin
                if      (hwdata == PASS_CODE) begin
                    c_led <= 4'b0101;
                end
                else if (hwdata == FAIL_CODE) begin
                    c_led <= 4'b1101;
                end
            end
       end
    end
endmodule
```

setting.xdc

```
## This file is a general .xdc for the Zybo Z7 Rev. B
## It is compatible with the Zybo Z7-20 and Zybo Z7-10
## To use it in a project:
## - uncomment the lines corresponding to used pins
```

```
## - rename the used ports (in each line, after get_ports) according to the top level
signal names in the project
# ports used: clk, led, ja
#Clock signal
set_property -dict { PACKAGE_PIN K17   IOSTANDARD LVCMOS33 } [get_ports { clk }];
#IO_L12P_T1_MRCC_35 Sch=sysclk
create_clock -add -name sys_clk_pin -period 20.00 -waveform {0 10} [get_ports { clk }];
##Switches
#set_property -dict { PACKAGE_PIN G15   IOSTANDARD LVCMOS33 } [get_ports { sw[0] }];
#IO_L19N_T3_VREF_35 Sch=sw[0]
#set_property -dict { PACKAGE_PIN P15   IOSTANDARD LVCMOS33 } [get_ports { sw[1] }];
#IO_L24P_T3_34 Sch=sw[1]
#set_property -dict { PACKAGE_PIN W13   IOSTANDARD LVCMOS33 } [get_ports { sw[2] }];
#IO_L4N_T0_34 Sch=sw[2]
#set_property -dict { PACKAGE_PIN T16   IOSTANDARD LVCMOS33 } [get_ports { sw[3] }];
#IO_L9P_T1_DQS_34 Sch=sw[3]
##Buttons
set_property -dict { PACKAGE_PIN K18   IOSTANDARD LVCMOS33 } [get_ports { rst_n }];
#IO_L12N_T1_MRCC_35 Sch=btn[0]
#set_property -dict { PACKAGE_PIN P16   IOSTANDARD LVCMOS33 } [get_ports { btn[1] }];
#IO_L24N_T3_34 Sch=btn[1]
#set_property -dict { PACKAGE_PIN K19   IOSTANDARD LVCMOS33 } [get_ports { btn[2] }];
#IO_L10P_T1_AD11P_35 Sch=btn[2]
#set_property -dict { PACKAGE_PIN Y16   IOSTANDARD LVCMOS33 } [get_ports { btn[3] }];
#IO_L7P_T1_34 Sch=btn[3]
#LEDs
set_property -dict { PACKAGE_PIN M14   IOSTANDARD LVCMOS33 } [get_ports { led[0] }];
#IO_L23P_T3_35 Sch=led[0]
set_property -dict { PACKAGE_PIN M15   IOSTANDARD LVCMOS33 } [get_ports { led[1] }];
#IO_L23N_T3_35 Sch=led[1]
set_property -dict { PACKAGE_PIN G14   IOSTANDARD LVCMOS33 } [get_ports { led[2] }];
#IO_0_35 Sch=led[2]
set_property -dict { PACKAGE_PIN D18   IOSTANDARD LVCMOS33 } [get_ports { led[3] }];
#IO_L3N_T0_DQS_AD1N_35 Sch=led[3]
##RGB LED 5 (Zybo Z7-20 only)
#set_property -dict { PACKAGE_PIN Y11   IOSTANDARD LVCMOS33 } [get_ports { led5_r }];
#IO_L18N_T2_13 Sch=led5_r
#set_property -dict { PACKAGE_PIN T5    IOSTANDARD LVCMOS33 } [get_ports { led5_g }];
#IO_L19P_T3_13 Sch=led5_g
#set_property -dict { PACKAGE_PIN Y12   IOSTANDARD LVCMOS33 } [get_ports { led5_b }];
#IO_L20P_T3_13 Sch=led5_b
##RGB LED 6
#set_property -dict { PACKAGE_PIN V16   IOSTANDARD LVCMOS33 } [get_ports { led6_r }];
#IO_L18P_T2_34 Sch=led6_r
#set_property -dict { PACKAGE_PIN F17   IOSTANDARD LVCMOS33 } [get_ports { led6_g }];
#IO_L6N_T0_VREF_35 Sch=led6_g
#set_property -dict { PACKAGE_PIN M17   IOSTANDARD LVCMOS33 } [get_ports { led6_b }];
#IO_L8P_T1_AD10P_35 Sch=led6_b
##Audio Codec
#set_property -dict { PACKAGE_PIN R19   IOSTANDARD LVCMOS33 } [get_ports { ac_bclk }];
#IO_0_34 Sch=ac_bclk
#set_property -dict { PACKAGE_PIN R17   IOSTANDARD LVCMOS33 } [get_ports { ac_mclk }];
#IO_L19N_T3_VREF_34 Sch=ac_mclk
#set_property -dict { PACKAGE_PIN P18   IOSTANDARD LVCMOS33 } [get_ports { ac_muten }];
#IO_L23N_T3_34 Sch=ac_muten
#set_property -dict { PACKAGE_PIN R18   IOSTANDARD LVCMOS33 } [get_ports { ac_pbdat }];
#IO_L20N_T3_34 Sch=ac_pbdat
#set_property -dict { PACKAGE_PIN T19   IOSTANDARD LVCMOS33 } [get_ports { ac_pblrc }];
#IO_25_34 Sch=ac_pblrc
#set_property -dict { PACKAGE_PIN R16   IOSTANDARD LVCMOS33 } [get_ports { ac_recdat }];
#IO_L19P_T3_34 Sch=ac_recdat
#set_property -dict { PACKAGE_PIN Y18   IOSTANDARD LVCMOS33 } [get_ports { ac_reclrc }];
#IO_L17P_T2_34 Sch=ac_reclrc
#set_property -dict { PACKAGE_PIN N18   IOSTANDARD LVCMOS33 } [get_ports { ac_scl }];
#IO_L13P_T2_MRCC_34 Sch=ac_scl
```

```
#set_property -dict { PACKAGE_PIN N17   IOSTANDARD LVCMOS33 } [get_ports { ac_sda }];
#IO_L23P_T3_34 Sch=ac_sda


##Additional Ethernet signals
#set_property -dict { PACKAGE_PIN F16   IOSTANDARD LVCMOS33  PULLUP true    } [get_ports
{ eth_int_pu_b }]; #IO_L6P_T0_35 Sch=eth_int_pu_b
#set_property -dict { PACKAGE_PIN E17   IOSTANDARD LVCMOS33 } [get_ports { eth_rst_b }];
#IO_L3P_T0_DQS_AD1P_35 Sch=eth_rst_b
##USB-OTG over-current detect pin
#set_property -dict { PACKAGE_PIN U13   IOSTANDARD LVCMOS33 } [get_ports { otg_oc }];
#IO_L3P_T0_DQS_PUDC_B_34 Sch=otg_oc
##Fan (Zybo Z7-20 only)
#set_property -dict { PACKAGE_PIN Y13   IOSTANDARD LVCMOS33  PULLUP true    } [get_ports
{ fan_fb_pu }]; #IO_L20N_T3_13 Sch=fan_fb_pu
##HDMI RX
#set_property -dict { PACKAGE_PIN W19   IOSTANDARD LVCMOS33 } [get_ports
{ hdmi_rx_hpd }]; #IO_L22N_T3_34 Sch=hdmi_rx_hpd
#set_property -dict { PACKAGE_PIN W18   IOSTANDARD LVCMOS33 } [get_ports
{ hdmi_rx_scl }]; #IO_L22P_T3_34 Sch=hdmi_rx_scl
#set_property -dict { PACKAGE_PIN Y19   IOSTANDARD LVCMOS33 } [get_ports
{ hdmi_rx_sda }]; #IO_L17N_T2_34 Sch=hdmi_rx_sda
#set_property -dict { PACKAGE_PIN U19   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_rx_clk_n }]; #IO_L12N_T1_MRCC_34 Sch=hdmi_rx_clk_n
#set_property -dict { PACKAGE_PIN U18   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_rx_clk_p }]; #IO_L12P_T1_MRCC_34 Sch=hdmi_rx_clk_p
#set_property -dict { PACKAGE_PIN W20   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_rx_n[0] }]; #IO_L16N_T2_34 Sch=hdmi_rx_n[0]
#set_property -dict { PACKAGE_PIN V20   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_rx_p[0] }]; #IO_L16P_T2_34 Sch=hdmi_rx_p[0]
#set_property -dict { PACKAGE_PIN U20   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_rx_n[1] }]; #IO_L15N_T2_DQS_34 Sch=hdmi_rx_n[1]
#set_property -dict { PACKAGE_PIN T20   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_rx_p[1] }]; #IO_L15P_T2_DQS_34 Sch=hdmi_rx_p[1]
#set_property -dict { PACKAGE_PIN P20   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_rx_n[2] }]; #IO_L14N_T2_SRCC_34 Sch=hdmi_rx_n[2]
#set_property -dict { PACKAGE_PIN N20   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_rx_p[2] }]; #IO_L14P_T2_SRCC_34 Sch=hdmi_rx_p[2]
##HDMI RX CEC (Zybo Z7-20 only)
#set_property -dict { PACKAGE_PIN Y8    IOSTANDARD LVCMOS33 } [get_ports
{ hdmi_rx_cec }]; #IO_L14N_T2_SRCC_13 Sch=hdmi_rx_cec
##HDMI TX
#set_property -dict { PACKAGE_PIN E18   IOSTANDARD LVCMOS33 } [get_ports
{ hdmi_tx_hpd }]; #IO_L5P_T0_AD9P_35 Sch=hdmi_tx_hpd
#set_property -dict { PACKAGE_PIN G17   IOSTANDARD LVCMOS33 } [get_ports
{ hdmi_tx_scl }]; #IO_L16P_T2_35 Sch=hdmi_tx_scl
#set_property -dict { PACKAGE_PIN G18   IOSTANDARD LVCMOS33 } [get_ports
{ hdmi_tx_sda }]; #IO_L16N_T2_35 Sch=hdmi_tx_sda
#set_property -dict { PACKAGE_PIN H17   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_tx_clk_n }]; #IO_L13N_T2_MRCC_35 Sch=hdmi_tx_clk_n
#set_property -dict { PACKAGE_PIN H16   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_tx_clk_p }]; #IO_L13P_T2_MRCC_35 Sch=hdmi_tx_clk_p
#set_property -dict { PACKAGE_PIN D20   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_tx_n[0] }]; #IO_L4N_T0_35 Sch=hdmi_tx_n[0]
#set_property -dict { PACKAGE_PIN D19   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_tx_p[0] }]; #IO_L4P_T0_35 Sch=hdmi_tx_p[0]
#set_property -dict { PACKAGE_PIN B20   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_tx_n[1] }]; #IO_L1N_T0_AD0N_35 Sch=hdmi_tx_n[1]
#set_property -dict { PACKAGE_PIN C20   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_tx_p[1] }]; #IO_L1P_T0_AD0P_35 Sch=hdmi_tx_p[1]
#set_property -dict { PACKAGE_PIN A20   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_tx_n[2] }]; #IO_L2N_T0_AD8N_35 Sch=hdmi_tx_n[2]
#set_property -dict { PACKAGE_PIN B19   IOSTANDARD TMDS_33     } [get_ports
{ hdmi_tx_p[2] }]; #IO_L2P_T0_AD8P_35 Sch=hdmi_tx_p[2]
##HDMI TX CEC
```

```
#set_property -dict { PACKAGE_PIN E19   IOSTANDARD LVCMOS33 } [get_ports
{ hdmi_tx_cec }]; #IO_L5N_T0_AD9N_35 Sch=hdmi_tx_cec

#Pmod Header JA (XADC)
#set_property -dict { PACKAGE_PIN N15   IOSTANDARD LVCMOS33 } [get_ports { ja[0] }];
#IO_L21P_T3_DQS_AD14P_35 Sch=JA1_R_p
#set_property -dict { PACKAGE_PIN L14   IOSTANDARD LVCMOS33 } [get_ports { ja[1] }];
#IO_L22P_T3_AD7P_35 Sch=JA2_R_P
#set_property -dict { PACKAGE_PIN K16   IOSTANDARD LVCMOS33 } [get_ports { ja[2] }];
#IO_L24P_T3_AD15P_35 Sch=JA3_R_P
#set_property -dict { PACKAGE_PIN K14   IOSTANDARD LVCMOS33 } [get_ports { ja[3] }];
#IO_L20P_T3_AD6P_35 Sch=JA4_R_P
#set_property -dict { PACKAGE_PIN N16   IOSTANDARD LVCMOS33 } [get_ports { ja[4] }];
#IO_L21N_T3_DQS_AD14N_35 Sch=JA1_R_N
#set_property -dict { PACKAGE_PIN L15   IOSTANDARD LVCMOS33 } [get_ports { ja[5] }];
#IO_L22N_T3_AD7N_35 Sch=JA2_R_N
#set_property -dict { PACKAGE_PIN J16   IOSTANDARD LVCMOS33 } [get_ports { ja[6] }];
#IO_L24N_T3_AD15N_35 Sch=JA3_R_N
#set_property -dict { PACKAGE_PIN J14   IOSTANDARD LVCMOS33 } [get_ports { ja[7] }];
#IO_L20N_T3_AD6N_35 Sch=JA4_R_N

##Pmod Header JB (Zybo Z7-20 only)
#set_property -dict { PACKAGE_PIN V8    IOSTANDARD LVCMOS33     } [get_ports { jb[0] }];
#IO_L15P_T2_DQS_13 Sch=jb_p[1]
#set_property -dict { PACKAGE_PIN W8    IOSTANDARD LVCMOS33     } [get_ports { jb[1] }];
#IO_L15N_T2_DQS_13 Sch=jb_n[1]
#set_property -dict { PACKAGE_PIN U7    IOSTANDARD LVCMOS33     } [get_ports { jb[2] }];
#IO_L11P_T1_SRCC_13 Sch=jb_p[2]
#set_property -dict { PACKAGE_PIN V7    IOSTANDARD LVCMOS33     } [get_ports { jb[3] }];
#IO_L11N_T1_SRCC_13 Sch=jb_n[2]
#set_property -dict { PACKAGE_PIN Y7    IOSTANDARD LVCMOS33     } [get_ports { jb[4] }];
#IO_L13P_T2_MRCC_13 Sch=jb_p[3]
#set_property -dict { PACKAGE_PIN Y6    IOSTANDARD LVCMOS33     } [get_ports { jb[5] }];
#IO_L13N_T2_MRCC_13 Sch=jb_n[3]
#set_property -dict { PACKAGE_PIN V6    IOSTANDARD LVCMOS33     } [get_ports { jb[6] }];
#IO_L22P_T3_13 Sch=jb_p[4]
#set_property -dict { PACKAGE_PIN W6    IOSTANDARD LVCMOS33     } [get_ports { jb[7] }];
#IO_L22N_T3_13 Sch=jb_n[4]


##Pmod Header JC
#set_property -dict { PACKAGE_PIN V15   IOSTANDARD LVCMOS33     } [get_ports { jc[0] }];
#IO_L10P_T1_34 Sch=jc_p[1]
#set_property -dict { PACKAGE_PIN W15   IOSTANDARD LVCMOS33     } [get_ports { jc[1] }];
#IO_L10N_T1_34 Sch=jc_n[1]
#set_property -dict { PACKAGE_PIN T11   IOSTANDARD LVCMOS33     } [get_ports { jc[2] }];
#IO_L1P_T0_34 Sch=jc_p[2]
#set_property -dict { PACKAGE_PIN T10   IOSTANDARD LVCMOS33     } [get_ports { jc[3] }];
#IO_L1N_T0_34 Sch=jc_n[2]
#set_property -dict { PACKAGE_PIN W14   IOSTANDARD LVCMOS33     } [get_ports { jc[4] }];
#IO_L8P_T1_34 Sch=jc_p[3]
#set_property -dict { PACKAGE_PIN Y14   IOSTANDARD LVCMOS33     } [get_ports { jc[5] }];
#IO_L8N_T1_34 Sch=jc_n[3]
#set_property -dict { PACKAGE_PIN T12   IOSTANDARD LVCMOS33     } [get_ports { jc[6] }];
#IO_L2P_T0_34 Sch=jc_p[4]
#set_property -dict { PACKAGE_PIN U12   IOSTANDARD LVCMOS33     } [get_ports { jc[7] }];
#IO_L2N_T0_34 Sch=jc_n[4]


##Pmod Header JD
#set_property -dict { PACKAGE_PIN T14   IOSTANDARD LVCMOS33     } [get_ports { jd[0] }];
#IO_L5P_T0_34 Sch=jd_p[1]
#set_property -dict { PACKAGE_PIN T15   IOSTANDARD LVCMOS33     } [get_ports { jd[1] }];
#IO_L5N_T0_34 Sch=jd_n[1]
#set_property -dict { PACKAGE_PIN P14   IOSTANDARD LVCMOS33     } [get_ports { jd[2] }];
#IO_L6P_T0_34 Sch=jd_p[2]
```

```
#set_property -dict { PACKAGE_PIN R14   IOSTANDARD LVCMOS33     } [get_ports { jd[3] }];
#IO_L6N_T0_VREF_34 Sch=jd_n[2]
#set_property -dict { PACKAGE_PIN U14   IOSTANDARD LVCMOS33     } [get_ports { jd[4] }];
#IO_L11P_T1_SRCC_34 Sch=jd_p[3]
#set_property -dict { PACKAGE_PIN U15   IOSTANDARD LVCMOS33     } [get_ports { jd[5] }];
#IO_L11N_T1_SRCC_34 Sch=jd_n[3]
#set_property -dict { PACKAGE_PIN V17   IOSTANDARD LVCMOS33     } [get_ports { jd[6] }];
#IO_L21P_T3_DQS_34 Sch=jd_p[4]
#set_property -dict { PACKAGE_PIN V18   IOSTANDARD LVCMOS33     } [get_ports { jd[7] }];
#IO_L21N_T3_DQS_34 Sch=jd_n[4]


##Pmod Header JE
#set_property -dict { PACKAGE_PIN V12   IOSTANDARD LVCMOS33 } [get_ports { je[0] }];
#IO_L4P_T0_34 Sch=je[1]
#set_property -dict { PACKAGE_PIN W16   IOSTANDARD LVCMOS33 } [get_ports { je[1] }];
#IO_L18N_T2_34 Sch=je[2]
#set_property -dict { PACKAGE_PIN J15   IOSTANDARD LVCMOS33 } [get_ports { je[2] }];
#IO_25_35 Sch=je[3]
#set_property -dict { PACKAGE_PIN H15   IOSTANDARD LVCMOS33 } [get_ports { je[3] }];
#IO_L19P_T3_35 Sch=je[4]
#set_property -dict { PACKAGE_PIN V13   IOSTANDARD LVCMOS33 } [get_ports { je[4] }];
#IO_L3N_T0_DQS_34 Sch=je[7]
#set_property -dict { PACKAGE_PIN U17   IOSTANDARD LVCMOS33 } [get_ports { je[5] }];
#IO_L9N_T1_DQS_34 Sch=je[8]
#set_property -dict { PACKAGE_PIN T17   IOSTANDARD LVCMOS33 } [get_ports { je[6] }];
#IO_L20P_T3_34 Sch=je[9]
#set_property -dict { PACKAGE_PIN Y17   IOSTANDARD LVCMOS33 } [get_ports { je[7] }];
#IO_L7N_T1_34 Sch=je[10]
##Pcam MIPI CSI-2 Connector
## This configuration expects the sensor to use 672Mbps/lane = 336 MHz HS_Clk
#create_clock -period 2.976 -name dphy_hs_clock_clk_p -waveform {0.000 1.488} [get_ports
dphy_hs_clock_clk_p]
#set_property INTERNAL_VREF 0.6 [get_iobanks 35]
#set_property -dict { PACKAGE_PIN J19   IOSTANDARD HSUL_12     } [get_ports
{ dphy_clk_lp_n }]; #IO_L10N_T1_AD11N_35 Sch=lp_clk_n
#set_property -dict { PACKAGE_PIN H20   IOSTANDARD HSUL_12     } [get_ports
{ dphy_clk_lp_p }]; #IO_L17N_T2_AD5N_35 Sch=lp_clk_p
#set_property -dict { PACKAGE_PIN M18   IOSTANDARD HSUL_12     } [get_ports
{ dphy_data_lp_n[0] }]; #IO_L8N_T1_AD10N_35 Sch=lp_lane_n[0]
#set_property -dict { PACKAGE_PIN L19   IOSTANDARD HSUL_12     } [get_ports
{ dphy_data_lp_p[0] }]; #IO_L9P_T1_DQS_AD3P_35 Sch=lp_lane_p[0]
#set_property -dict { PACKAGE_PIN L20   IOSTANDARD HSUL_12     } [get_ports
{ dphy_data_lp_n[1] }]; #IO_L9N_T1_DQS_AD3N_35 Sch=lp_lane_n[1]
#set_property -dict { PACKAGE_PIN J20   IOSTANDARD HSUL_12     } [get_ports
{ dphy_data_lp_p[1] }]; #IO_L17P_T2_AD5P_35 Sch=lp_lane_p[1]
#set_property -dict { PACKAGE_PIN H18   IOSTANDARD LVDS_25     } [get_ports
{ dphy_hs_clock_clk_n }]; #IO_L14N_T2_AD4N_SRCC_35 Sch=mipi_clk_n
#set_property -dict { PACKAGE_PIN J18   IOSTANDARD LVDS_25     } [get_ports
{ dphy_hs_clock_clk_p }]; #IO_L14P_T2_AD4P_SRCC_35 Sch=mipi_clk_p
#set_property -dict { PACKAGE_PIN M20   IOSTANDARD LVDS_25     } [get_ports
{ dphy_data_hs_n[0] }]; #IO_L7N_T1_AD2N_35 Sch=mipi_lane_n[0]
#set_property -dict { PACKAGE_PIN M19   IOSTANDARD LVDS_25     } [get_ports
{ dphy_data_hs_p[0] }]; #IO_L7P_T1_AD2P_35 Sch=mipi_lane_p[0]
#set_property -dict { PACKAGE_PIN L17   IOSTANDARD LVDS_25     } [get_ports
{ dphy_data_hs_n[1] }]; #IO_L11N_T1_SRCC_35 Sch=mipi_lane_n[1]
#set_property -dict { PACKAGE_PIN L16   IOSTANDARD LVDS_25     } [get_ports
{ dphy_data_hs_p[1] }]; #IO_L11P_T1_SRCC_35 Sch=mipi_lane_p[1]
#set_property -dict { PACKAGE_PIN G19   IOSTANDARD LVCMOS33 } [get_ports { cam_clk }];
#IO_L18P_T2_AD13P_35 Sch=cam_clk
#set_property -dict { PACKAGE_PIN G20   IOSTANDARD LVCMOS33      PULLUP true} [get_ports
{ cam_gpio }]; #IO_L18N_T2_AD13N_35 Sch=cam_gpio
#set_property -dict { PACKAGE_PIN F20   IOSTANDARD LVCMOS33 } [get_ports { cam_scl }];
#IO_L15N_T2_DQS_AD12N_35 Sch=cam_scl
#set_property -dict { PACKAGE_PIN F19   IOSTANDARD LVCMOS33 } [get_ports { cam_sda }];
#IO_L15P_T2_DQS_AD12P_35 Sch=cam_sda
```

```
##Unloaded Crypto Chip SWI (for future use)
#set_property -dict { PACKAGE_PIN P19   IOSTANDARD LVCMOS33 } [get_ports
{ crypto_sda }]; #IO_L13N_T2_MRCC_34 Sch=crypto_sda


##Unconnected Pins (Zybo Z7-20 only)
#set_property PACKAGE_PIN T9 [get_ports {netic19_t9}]; #IO_L12P_T1_MRCC_13
#set_property PACKAGE_PIN U10 [get_ports {netic19_u10}]; #IO_L12N_T1_MRCC_13
#set_property PACKAGE_PIN U5 [get_ports {netic19_u5}]; #IO_L19N_T3_VREF_13
#set_property PACKAGE_PIN U8 [get_ports {netic19_u8}]; #IO_L17N_T2_13
#set_property PACKAGE_PIN U9 [get_ports {netic19_u9}]; #IO_L17P_T2_13
#set_property PACKAGE_PIN V10 [get_ports {netic19_v10}]; #IO_L21N_T3_DQS_13
#set_property PACKAGE_PIN V11 [get_ports {netic19_v11}]; #IO_L21P_T3_DQS_13
#set_property PACKAGE_PIN V5 [get_ports {netic19_v5}]; #IO_L6N_T0_VREF_13
#set_property PACKAGE_PIN W10 [get_ports {netic19_w10}]; #IO_L16P_T2_13
#set_property PACKAGE_PIN W11 [get_ports {netic19_w11}]; #IO_L18P_T2_13
#set_property PACKAGE_PIN W9 [get_ports {netic19_w9}]; #IO_L16N_T2_13
#set_property PACKAGE_PIN Y9 [get_ports {netic19_y9}]; #IO_L14P_T2_SRCC_13
```